



Branding is what people say about you when you are not in the room.



-Jeff Bezos



Delhi Police's proposed ITMS

ITMS approved by the government of India for Delhi is largely based on AI to substantially reduce human intervention in the city's day-to-day traffic management. The project includes a traffic control and management centre, a disaster-recovery centre, an incident-detection system on expressways and adaptive signal control of intersections.

It will also involve a CCTV camera system, traffic information system, adaptive speed control on national highways, speed- and traffic-light-violation cameras, and a parking-management system. Road-based laser sensors will help in traffic classification and traffic density count, and generate a pattern of traffic movement based on traffic volume.

An adaptive traffic-control system will control traffic movement at intersections on arterial roads that usually clog up during peak hours. Road-based sensors will constantly monitor the flow of traffic and provide live data to the control room. When traffic on a stretch becomes heavy, the control room will divert vehicles approaching the congestion using variable message boards and IP-based public-address systems.

Computers monitoring the traffic will leave signals green on a vehicle-heavy stretch for a longer period to clear bottlenecks. Sensors will also anticipate traffic congestion, take pre-emptive measures and post information on Facebook, Twitter, Google Map and other social network platforms about traffic snarls and road blocks.

The system will generate data relating to traffic movement in the context of time of day and weather condition for future reference. Data will then be stored in a cloud-based system accessible to police officials through a single software. The project will enable faster and efficient traffic management in case of an emergency, such as an accident, fire or a blockade.

An automatic system will be enabled by a series of cameras equipped with high-resolution number-plate-detection lenses to determine stop-line violations, lane violations, speeding, travelling without seat belts or use of a cellphone while driving. Once detected, e-challan systems will send a notice to the violator's address, and alert the violator on his or her registered cell number.

Video incident-detection cameras, proactive variable message signage and speed-control signals will help in reducing the incidence of accidents on roads.

SMART MANAGING OF TRAFFIC

Traffic Safety Management System will come up between Sarai Kale Khan and Dhaula Kuan, covering 10 junctions

TRIAL RUN



THE OBJECTIVES

Reduction in accidents, violations | Tighter law enforcement through e-challans | Surveillance

Advanced cameras can detect signal and speed violations, numberplates, and send data to single control room

Delhi's proposed ITMS (Credit: <https://timesofindia.indiatimes.com>)

information management system, communication sub-system, travel demand management, incident and emergency management system, and fleet management system.

The Centre for Development of Advanced Greater Hyderabad has a master plan for implementing ITMS soon with an objective to support optimisation of traffic flow, reduce environmental impact, bring down road accidents, and promote efficiency in traffic and transportation operation and management.

There are further projects in the pipeline across India, such as implementation of ITMS in Indore, Pune, etc.

Conclusion

ITMS is a useful innovation that provides a solution to the existing framework of inadequate traffic management system and its related Web structure (like toll deduction, chal-

lan due to over speeding, pollution checking service, VIP/ambulance/fire tender path clearance, vehicle theft and so on).

All these issues are efficiently managed by ITMS to tackle the menace of existing traffic mis-management. Success of the implemented ITMS requires strong backup in terms of policy, legislation, regulation, relevant sector coordination and common action.

ITMS can fail to deliver the expected benefits due to failure not of technology elements, but of some institutional elements. Such systems have been implemented with great success on the fast roads of developed countries like the US, Japan and the UK. Moreover, various researchers are working to make these traffic systems even more advanced. While India has started taking steps in planning and implementation of ITMS, our country still has a long way to go! **EFY**